Included Variable Bias in Leipziger (2024): An Empirical Application

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1 Introduction

Leipziger (2024, AJPS) estimates the effect of democratic transitions on socioeconomic ethnic inequality using a two-way fixed effects (TWFE) model with country-year panel data. The main specification is:

$$\text{Ethnic Inequality}_{it} = \beta\,\text{Democracy}(0,1)_{i,t-1} + \theta\,X_{i,t-1} + \delta_i + \gamma_t + \varepsilon_{it}$$

where ethnic inequality is measured by three different indices (V-Dem public services, Alesina et al. nightlight income, Omoeva et al. education), Democracy is a binary indicator based on the Lexical Index (level ≥ 5), and $X_{i,t-1}$ is log GDP per capita (the only time-varying control in the baseline specification). Country fixed effects (δ_i) absorb time-invariant confounders, and year fixed effects (γ_t) absorb global shocks.

A key question is whether GDP per capita is a **collider** — a variable caused by both the treatment (democracy) and the outcome (ethnic inequality). If so, conditioning on it introduces **Included Variable Bias** (IVB):

$$\text{IVB}(\hat{\beta}) = \hat{\beta}^* - \hat{\beta} = -\hat{\theta}^* \cdot \hat{\pi}$$

where $\hat{\theta}^*$ is the coefficient of GDP per capita in the "long" model and $\hat{\pi}$ is the coefficient of democracy in the auxiliary regression of GDP per capita on democracy and other controls.

2 Data

```r
dat <- read.delim(
  "candidate_papers/leipziger_2024/Country-level_dataset.tab",
  stringsAsFactors = FALSE
)

# Generate variables
dat$lexical_index_5 <- ifelse(dat$lexical_index >= 5, 1,
                              ifelse(dat$lexical_index <= 4, 0, NA))

dat$SEI <- (dat$v2peapssoc - 3.37) / (-3.135 - 3.37)

dat <- dat %>%
  arrange(country_id, year) %>%
  group_by(country_id) %>%
  mutate(
    L_lexical_index_5 = dplyr::lag(lexical_index_5, 1),
    L_latent_gdppc_mean_log = dplyr::lag(latent_gdppc_mean_log, 1),
    L_e_total_oil_income_pc = dplyr::lag(e_total_oil_income_pc, 1),
```

```r
    L_e_civil_war = dplyr::lag(e_civil_war, 1),
    L_efindex = dplyr::lag(efindex, 1),
    L_e_migdpgro = dplyr::lag(e_migdpgro, 1)
  ) %>%
  ungroup()

dat$year_f <- factor(dat$year)
dat$country_f <- factor(dat$country_id)
```

The dataset contains 53410 country-year observations for 231 countries over 1789–2020.

3 Replication of Table 1

3.1 Baseline TWFE

```r
mod_sei <- feols(
  SEI ~ L_lexical_index_5 + L_latent_gdppc_mean_log | country_f + year_f,
  data = dat, vcov = ~country_id
)

mod_grg <- feols(
  grg ~ L_lexical_index_5 + L_latent_gdppc_mean_log | country_f + year_f,
  data = dat, vcov = ~country_id
)

mod_ggini <- feols(
  ggini ~ L_lexical_index_5 + L_latent_gdppc_mean_log | country_f + year_f,
  data = dat, vcov = ~country_id
)
```

```r
comparison <- data.frame(
  Outcome = c("Public services (SEI)", "Income (Alesina)", "Education (Omoeva)"),
  N_published = c(15188, 517, 3945),
  N_replicated = c(mod_sei$nobs, mod_grg$nobs, mod_ggini$nobs),
  Beta_published = c(-0.035, -0.030, -0.005),
  Beta_replicated = c(
    round(coef(mod_sei)["L_lexical_index_5"], 3),
    round(coef(mod_grg)["L_lexical_index_5"], 3),
    round(coef(mod_ggini)["L_lexical_index_5"], 3)
  )
)
kable(comparison, digits = 3,
      caption = "Table 1 Replication: Published vs. Replicated",
      col.names = c("Outcome", "N (pub.)", "N (rep.)", "$\\hat\\beta$ (pub.)",
                    "$\\hat\\beta$ (rep.)"))
```

Table 1: Table 1 Replication: Published vs. Replicated

Outcome	N (pub.)	N (rep.)	$\hat{\beta}$ (pub.)	$\hat{\beta}$ (rep.)
Public services (SEI)	15188	15191	-0.035	-0.035
Income (Alesina)	517	517	-0.030	-0.030
Education (Omoeva)	3945	3945	-0.005	-0.005

Coefficients match closely across all three outcomes.

4 IVB Decomposition

4.1 Setup

We define:

- **Treatment** (d): `L_lexical_index_5` (lagged binary democracy)
- **Outcome** (y): each of the three ethnic inequality measures
- **Candidate collider** (z): `L_latent_gdppc_mean_log` (lagged log GDP per capita)
- **Fixed effects**: country + year

The baseline model has **only one** time-varying control (GDP per capita), making the IVB decomposition straightforward: the "short" model excludes GDP per capita entirely (only FE remain), and the "long" model includes it.

4.2 Table 1: GDP per capita as candidate collider

```r
outcomes <- c("SEI", "grg", "ggini")
outcome_labels <- c("Public services (SEI)", "Income (Alesina)", "Education (Omoeva)")

ivb_table1 <- lapply(outcomes, function(y_var) {
  compute_ivb_multi(
    data = dat, y = y_var, d_vars = "L_lexical_index_5",
    z = "L_latent_gdppc_mean_log", w = character(),
    fe = c("country_f", "year_f"), vcov = "iid"
  )
})
names(ivb_table1) <- outcomes

tab_t1 <- do.call(rbind, lapply(seq_along(outcomes), function(i) {
  r <- ivb_table1[[outcomes[i]]]$results
  data.frame(
    Outcome = outcome_labels[i],
    N = ivb_table1[[outcomes[i]]]$sample_n,
    beta_short = r$beta_short,
    beta_long = r$beta_long,
    theta = r$theta,
    pi = r$pi,
    IVB = r$ivb_formula,
    check = r$diff_check,
    stringsAsFactors = FALSE
  )
}))
```

```r
kable(tab_t1, digits = 4, row.names = FALSE,
      caption = "IVB decomposition: GDP per capita as candidate collider (Table 1)",
      col.names = c("Outcome", "N", "$\\hat\\beta_{\\text{short}}$",
                    "$\\hat\\beta_{\\text{long}}$", "$\\hat\\theta^*$",
                    "$\\hat\\pi$", "IVB $= -\\hat\\theta^* \\hat\\pi$",
                    "Check"))
```

Table 2: IVB decomposition: GDP per capita as candidate collider (Table 1)

Outcome	N	$\hat{\beta}_{\text{short}}$	$\hat{\beta}_{\text{long}}$	$\hat{\theta}^*$	$\hat{\pi}$	IVB $= -\hat{\theta}^*\hat{\pi}$	Check
Public services (SEI)	15191	-0.0407	-0.0352	-0.0497	0.1123	0.0056	0
Income (Alesina)	517	-0.0295	-0.0298	-0.0227	-0.0119	-0.0003	0
Education (Omoeva)	3945	-0.0043	-0.0051	0.0126	0.0606	-0.0008	0

The identity holds exactly (check ≈ 0) in all cases. The key finding: for the **public services** outcome (the most robust and well-powered specification), including GDP per capita as a control produces an IVB of **+0.006**, which *attenuates* the estimated effect of democracy by 16% (from -0.041 to -0.035).

4.3 Extended Model (Table A11): Multiple controls

```
all_controls <- c("L_latent_gdppc_mean_log", "L_e_total_oil_income_pc",
                  "L_e_civil_war", "L_efindex", "L_e_migdpgro")
control_labels <- c("Log GDP p.c.", "Oil income p.c.", "Civil war",
                    "Ethnic frac.", "GDP growth")

ivb_ext <- lapply(seq_along(all_controls), function(j) {
  z_var <- all_controls[j]
  w_vars <- setdiff(all_controls, z_var)
  tryCatch(
    compute_ivb_multi(
      data = dat, y = "SEI", d_vars = "L_lexical_index_5",
      z = z_var, w = w_vars,
      fe = c("country_f", "year_f"), vcov = "iid"
    ),
    error = function(e) NULL
  )
})
names(ivb_ext) <- all_controls

tab_ext <- do.call(rbind, lapply(seq_along(all_controls), function(j) {
  if (is.null(ivb_ext[[j]])) return(NULL)
  r <- ivb_ext[[j]]$results
  data.frame(
    Collider = control_labels[j],
    beta_short = r$beta_short,
    beta_long = r$beta_long,
    theta = r$theta,
    pi = r$pi,
    IVB = r$ivb_formula,
    check = r$diff_check,
    stringsAsFactors = FALSE
  )
}))
```

```r
kable(tab_ext, digits = 4, row.names = FALSE,
      caption = "IVB decomposition for each control (Table A11, SEI outcome)",
      col.names = c("Candidate collider $z$", "$\\hat\\beta_{\\text{short}}$",
                    "$\\hat\\beta_{\\text{long}}$", "$\\hat\\theta^*$",
                    "$\\hat\\pi$", "IVB $= -\\hat\\theta^* \\hat\\pi$",
                    "Check"))
```

Table 3: IVB decomposition for each control (Table A11, SEI outcome)

Candidate collider z	$\hat{\beta}_{\text{short}}$	$\hat{\beta}_{\text{long}}$	$\hat{\theta}^*$	$\hat{\pi}$	IVB $= -\hat{\theta}^*\hat{\pi}$	Check
Log GDP p.c.	-0.0359	-0.0311	-0.0473	0.1012	0.0048	0
Oil income p.c.	-0.0311	-0.0311	0.0000	49.7449	0.0000	0
Civil war	-0.0314	-0.0311	0.0110	-0.0210	0.0002	0
Ethnic frac.	-0.0318	-0.0311	0.0463	-0.0143	0.0007	0
GDP growth	-0.0311	-0.0311	0.0216	0.0006	0.0000	0

GDP per capita produces the largest IVB (+0.005) even in the extended specification. All other controls contribute IVBs smaller than 0.001 in absolute value.

5 Literature Support for Collider Status

The IVB formula diagnoses **how much** each control shifts the treatment estimate, but **whether** a control is a collider depends on substantive knowledge: does the outcome (ethnic inequality) cause the control z?

5.1 Ethnic inequality → GDP per capita

There is moderate-to-strong evidence that ethnic inequality reduces economic growth and GDP:

- **Gründler & Link (2024, CESifo WP):** Using harmonized satellite data and IV (artificiality of subnational borders in Africa), find that ethnic inequality causally reduces GDP per capita by 12–15% at extreme levels.
- **Easterly & Levine (1997, QJE):** Ethnic fragmentation explains Africa's slow growth via bad policies, instability, and low schooling.
- **Alesina, Michalopoulos & Papaioannou (2016, JPE):** Strong inverse association between ethnic inequality (nightlight Gini across homelands) and development.

Implication: GDP per capita is a plausible collider. Ethnic inequality reduces GDP, and democracy increases GDP (Acemoglu et al. 2019). Including GDP per capita opens a non-causal path, attenuating the estimated effect of democracy.

5.2 Ethnic inequality → Civil war

There is strong evidence that horizontal inequalities cause civil war:

- **Cederman, Gleditsch & Buhaug (2013, Cambridge UP):** Politically excluded and economically disadvantaged ethnic groups face significantly higher conflict risk. Group-level analysis with EPR data.
- **Wucherpfennig et al. (2016, AJPS):** IV using colonial rule strategies (British vs. French) shows that ethnic exclusion *causally* increases conflict; naïve estimates *understate* the effect.
- **Stewart (2008, Palgrave):** Multi-dimensional HIs across economic, social, and political dimensions provoke conflict; strongest when dimensions reinforce each other.

Implication: Civil war is a plausible collider, though the empirical IVB is small (+0.0002) due to the modest conditional effect of civil war on ethnic inequality within this sample.

5.3 Controls unlikely to be colliders

- **Oil income per capita:** Determined by geological endowment and world prices; no plausible channel from ethnic inequality to oil revenues.
- **Ethnic fractionalization:** Structural/demographic variable measuring group composition; changes over decades/centuries, not affected by contemporaneous inequality.
- **GDP growth:** Same mechanisms as GDP per capita, but empirical IVB is negligible.

6 Summary

```r
summary_all <- data.frame(
  Control = c("Log GDP p.c.", "Oil income p.c.", "Civil war",
              "Ethnic frac.", "GDP growth"),
  `Y causes Z` = c("Moderate-strong", "No", "Strong",
                   "No (structural)", "Moderate"),
  `Key reference` = c("Gründler & Link 2024", "---",
                      "Cederman et al. 2013", "---", "Easterly & Levine 1997"),
  `IVB (Table 1)` = c(round(tab_t1$IVB[1], 4), rep(NA, 4)),
  `IVB (Table A11)` = round(tab_ext$IVB, 4),
  check.names = FALSE, stringsAsFactors = FALSE
)
kable(summary_all, caption = "Summary: Literature evidence and IVB magnitudes")
```

Table 4: Summary: Literature evidence and IVB magnitudes

Control	Y causes Z	Key reference	IVB (Table 1)	IVB (Table A11)
Log GDP p.c.	Moderate-strong	Gründler & Link 2024	0.0056	0.0048
Oil income p.c.	No	—	NA	0.0000
Civil war	Strong	Cederman et al. 2013	NA	0.0002
Ethnic frac.	No (structural)	—	NA	0.0007
GDP growth	Moderate	Easterly & Levine 1997	NA	0.0000

The main finding is that GDP per capita — the only time-varying control in Leipziger's baseline specification — is a plausible collider with a substantive IVB. Including it **attenuates** the estimated effect of democracy on ethnic inequality by 16% for the public services measure. Leipziger (2024) notes that additional controls are excluded due to "the risk of posttreatment bias" (p. 1347), which is consistent with IVB/collider bias concerns. However, the retained control (GDP per capita) introduces the largest bias of all. The "short" model without GDP per capita yields $\hat{\beta} = -0.041$, suggesting the true equalizing effect of democracy may be larger than the reported -0.035.

7 References

- Acemoglu, D., Naidu, S., Restrepo, P., & Robinson, J. A. (2019). Democracy Does Cause Growth. *Journal of Political Economy*, 127(1), 47–100.
- Alesina, A., Michalopoulos, S., & Papaioannou, E. (2016). Ethnic Inequality. *Journal of Political Economy*, 124(2), 428–488.
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